

Phonological Assembly Circuits in GPT-2

How Transformers Compute Sound Structure from Orthography

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Part 1: The Question

Why phonological circuits?

The ML literature has characterized circuits for:

- Syntax: IOI, subject–verb agreement, reflexive anaphora
- Retrieval: induction heads, greater-than
- Factual: capital cities, birth years

Gap: zero circuit-level work on **phonological** computation.

Yet LLMs demonstrably encode sound structure:

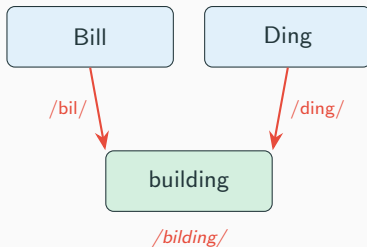
- Linear probes on GPT-2 mid-layers: 63–79% rhyme detection (Liao & Shi, 2026)
- Above-chance grapheme-to-phoneme conversion (PhonologyBench, 2024)

Question: *Which attention heads compute phonological knowledge?*

The puzzle

BPE tokenization encodes **orthography**, not phonology.

Yet the model can do this:



Cross-boundary phoneme fusion from purely orthographic input.

How? Which heads? Is there a dedicated circuit?

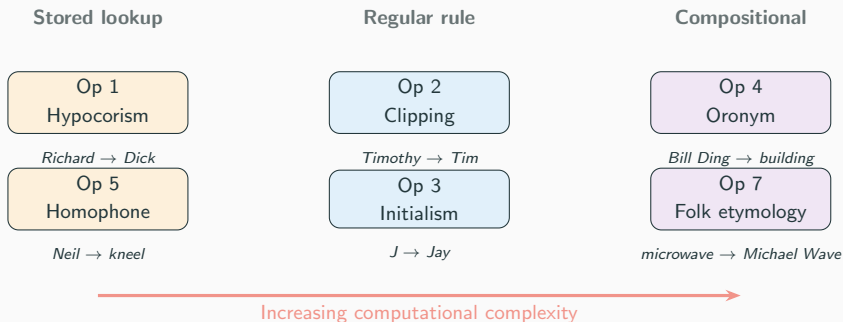
Six phonological operations

Op	Linguistic term	Example	Computation	N
1	Rhyming hypocorism	Richard → Dick	Irregular stored lookup	32
2	Clipping (apocope)	Timothy → Tim	First-syllable truncation	90
3	Initialism	J → Jay	Grapheme → phoneme name	30
4	Oronym	Bill Ding → building	Cross-boundary fusion	99
5	Homophone	Neil → kneel	Phoneme-identity detection	120
7	Folk etymology	microwave → Michael Wave	Reverse decomposition	75

Each requires a **linguistically distinct** computation.

⇒ Predicts **distinct circuit architectures**.

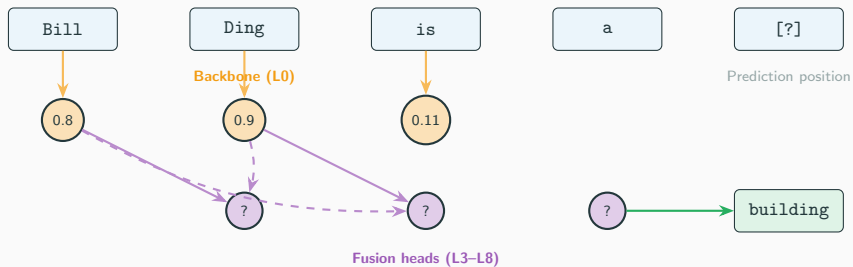
Operations as a computational taxonomy



Stored → Regular → Compositional: each category demands more computation.

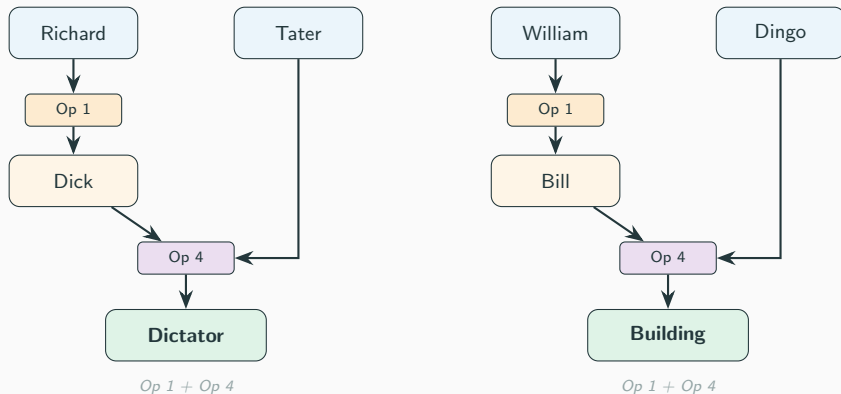
Op 4 (oronym) requires **cross-token phoneme assembly** — the hardest operation.

The oronym problem



Expected: fusion heads attend *across* the word boundary, composing phonemes from both “Bill” and “Ding” into “building.”

Compound puns: chaining operations



Compositionality test: ablate Op 1 circuit $\Rightarrow$ compound breaks, Op 4 survives.

Ablate Op 4 circuit $\Rightarrow$ compound breaks, Op 1 survives.

Part 2: Methods

Minimal-pair datasets

Each pair follows BLiMP format: {good, bad, target_token, foil_token}

Operation	Good prompt	Target / Foil
Op 1	"People call Richard by the name"	Dick / Mark
Op 2	"Timothy goes by"	Tim / Tom
Op 3	"The letter J is pronounced"	Jay / Gee
Op 4	"Bill Ding sounds like"	building / nothing
Op 5	"Neil sounds like"	kneel / wheel
Op 7	"Mike Rowave sounds like"	microwave / nothing

Evaluation: logit difference at final position.

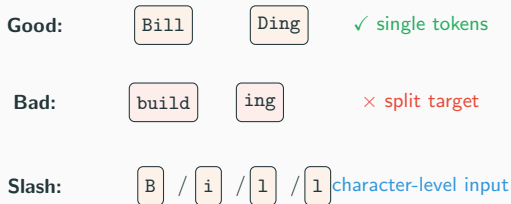
Gate: >70% accuracy before circuit discovery.

Ops 2, 3, 5 are procedurally expandable (CMU Dict).

Ops 1, 4, 7 are controlled vocabularies.

Tokenization audit

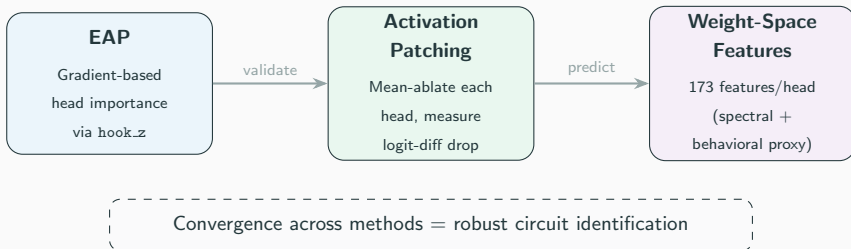
BPE tokenization can break phonological tasks:



Following Liao & Shi (2026): if slash-delimited input improves Op 4 accuracy, the circuit *exists* but is bottlenecked by tokenization.

⇒ Weight-space capacity vs. activation-flow distinction.

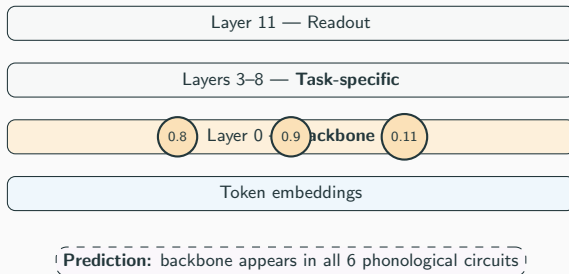
Three discovery methods



Part 3: Circuits

The backbone hypothesis

Prior finding: heads L0H8, L0H9, L0H11 participate in every known circuit (IOI, SVA, greater-than, induction).



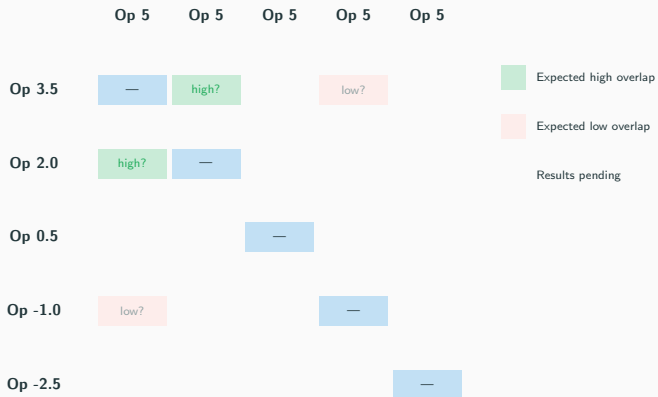
If confirmed: the “operating system” extends to phonological computation.

Predicted circuit architectures

Op	Expected circuit type	Key heads	Analogy
1	Associative lookup	Mid-layer OV	Name-mover (IOI)
2	Positional truncation	Positional attn	Acronym circuit
3	G2P map	Letter → phoneme	Letter-mover
4	Cross-boundary fusion	Novel L3–L8	None known
5	Phoneme identity	Mid-layer probe	Rhyme probes
7	Reverse decomposition	Creative gen	Expected to fail

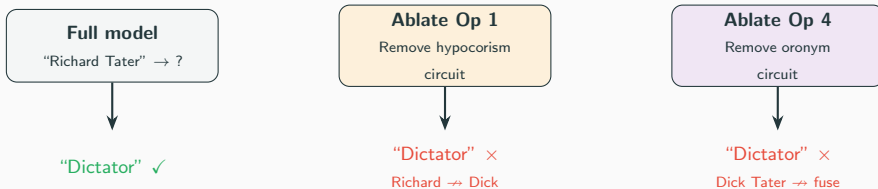
Op 4 is the core novel finding: if cross-boundary fusion heads exist, they are **a new circuit type** not identified in any prior work.

Circuit overlap matrix



If Op 4 uses *unique* heads not shared with any other operation:
evidence for specialized phonological circuitry.

Compositional validation protocol

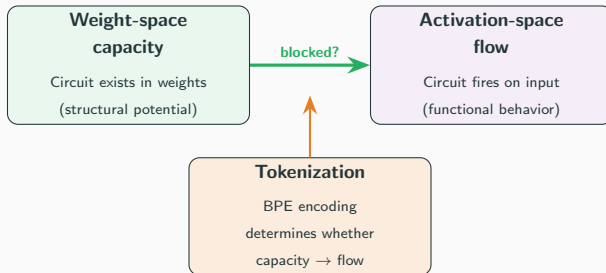


Each component circuit is **independently necessary**.

⇒ True compositional structure, not a single monolithic circuit.

Part 4: Implications

Tokenization as computational bottleneck



If Op 4 accuracy improves with character-level input (slash condition),
GPT-2 *has* the circuit but BPE *prevents it from firing*.

What this tells us

If the oronym circuit exists:

- GPT-2 learned cross-boundary phonological assembly from orthographic training alone
- Phonological computation is not a single undifferentiated capability — distinct operations use distinct circuits
- The backbone “operating system” extends to phonology

If it doesn't:

- Behavioral phonological performance (probes, PhonologyBench) may be statistical co-occurrence, not structured computation
- The weight-capacity vs. activation-flow gap is real and measurable

Either outcome is a publishable finding.

The dataset: selected examples

Input	Target	Operation chain
Richard Tater	Dictator	Op 1 + Op 4
William Ding	Building	Op 1 + Op 4
Elizabeth Aardvark	Lizard	Op 2 + Op 4
Bill Ding	Building	Op 4 only
Curt Ain	Curtain	Op 4 only
Brock Lee	Broccoli	Op 4 only
Barb Dwyer	Barbed wire	Op 4 only
Mike Rowave	Microwave	Op 7 (reverse)
Dan Druff	Dandruff	Op 4 only
Herb Ivore	Herbivore	Op 4 only

99 oronym pairs, 32 hypocorism pairs, 120 homophone pairs.

446 total minimal pairs across 6 operations.

Current status

Component	Status	Next step
Paper outline	Complete	—
6 operation definitions	Complete	—
Minimal-pair datasets	Designed	Build & audit tokenization
Behavioral validation	Pending	Run on GPT-2 S/M/L
EAP discovery	Pending	Run per-operation
Activation patching	Pending	Validate EAP rankings
Weight-space prediction	Pending	Cross-reference
Compositional ablation	Pending	After circuit ID
Tokenization experiment	Pending	Slash-condition test

Target venue: COLM, Findings of ACL, or MI workshop.

Phonological Assembly Circuits in GPT-2

Repository: github.com/elliotttower/phonetic-circuits

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“Bill Ding” → “building”